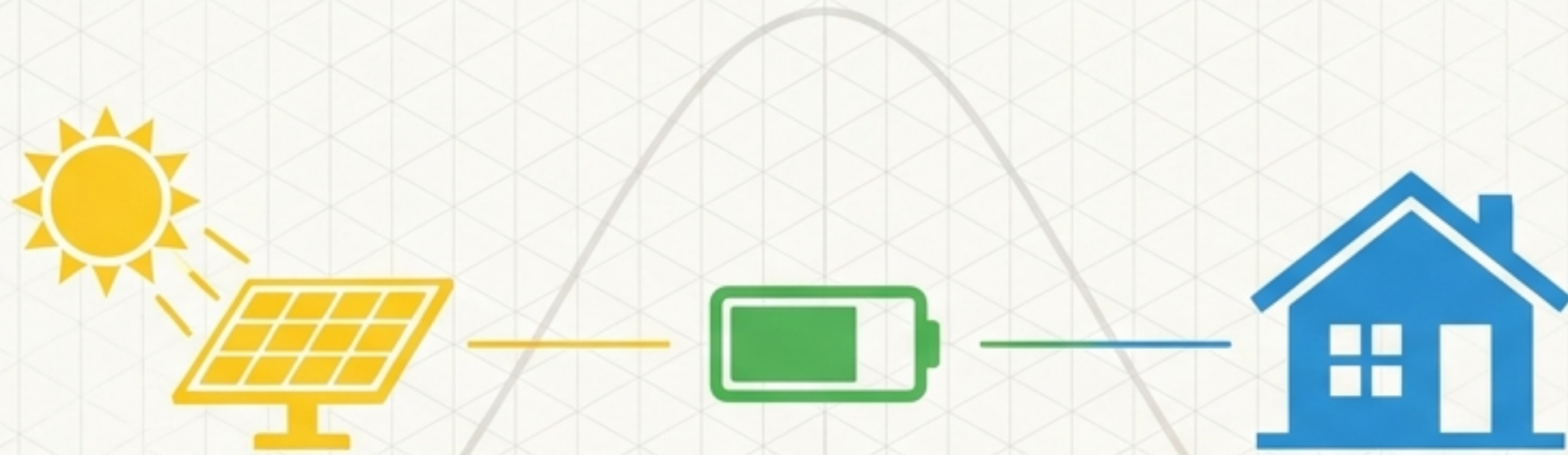
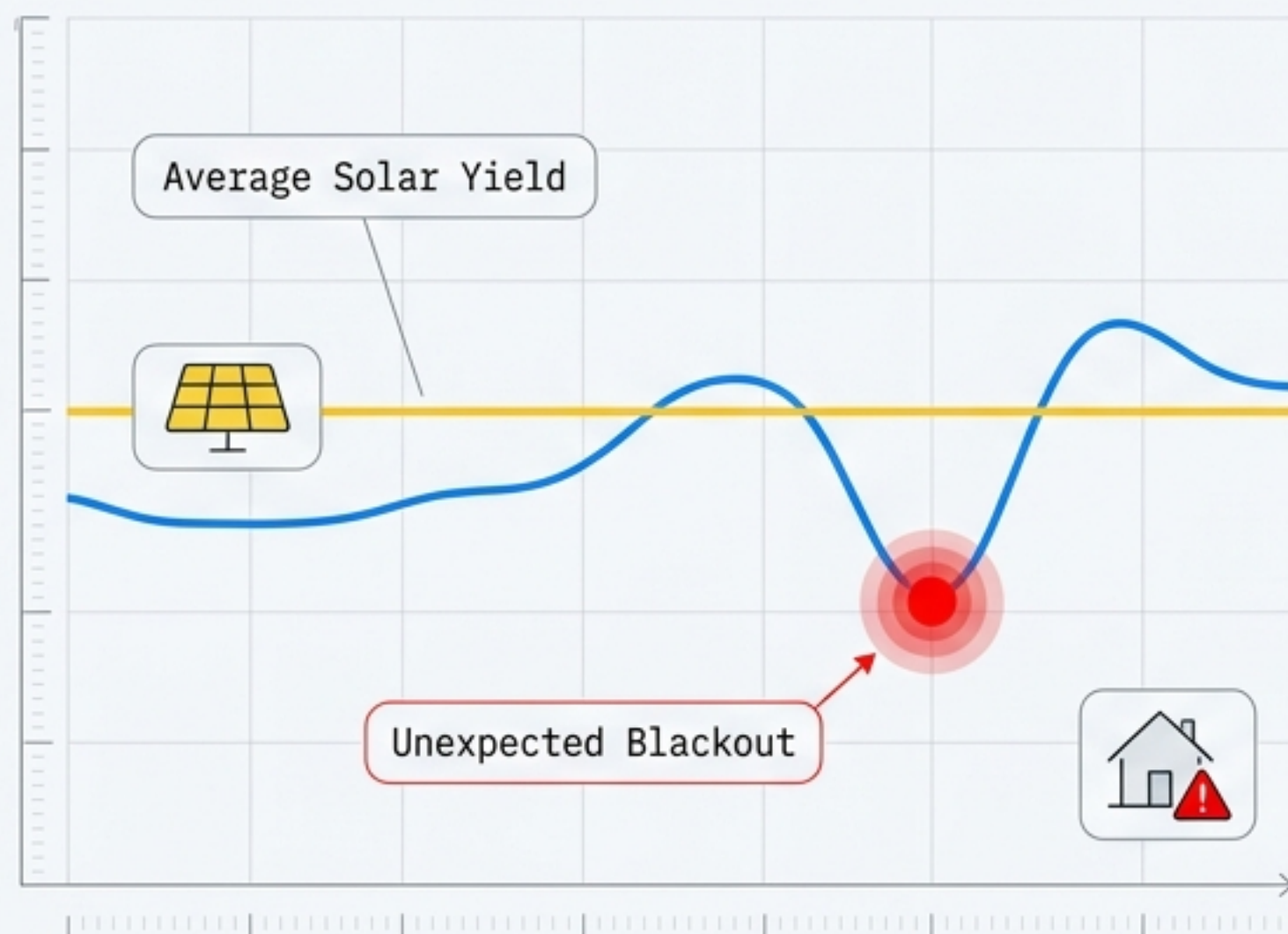


Navigating Uncertainty in Solar Energy

A Probabilistic Analysis & Monte Carlo Simulation of Off-Grid System Reliability



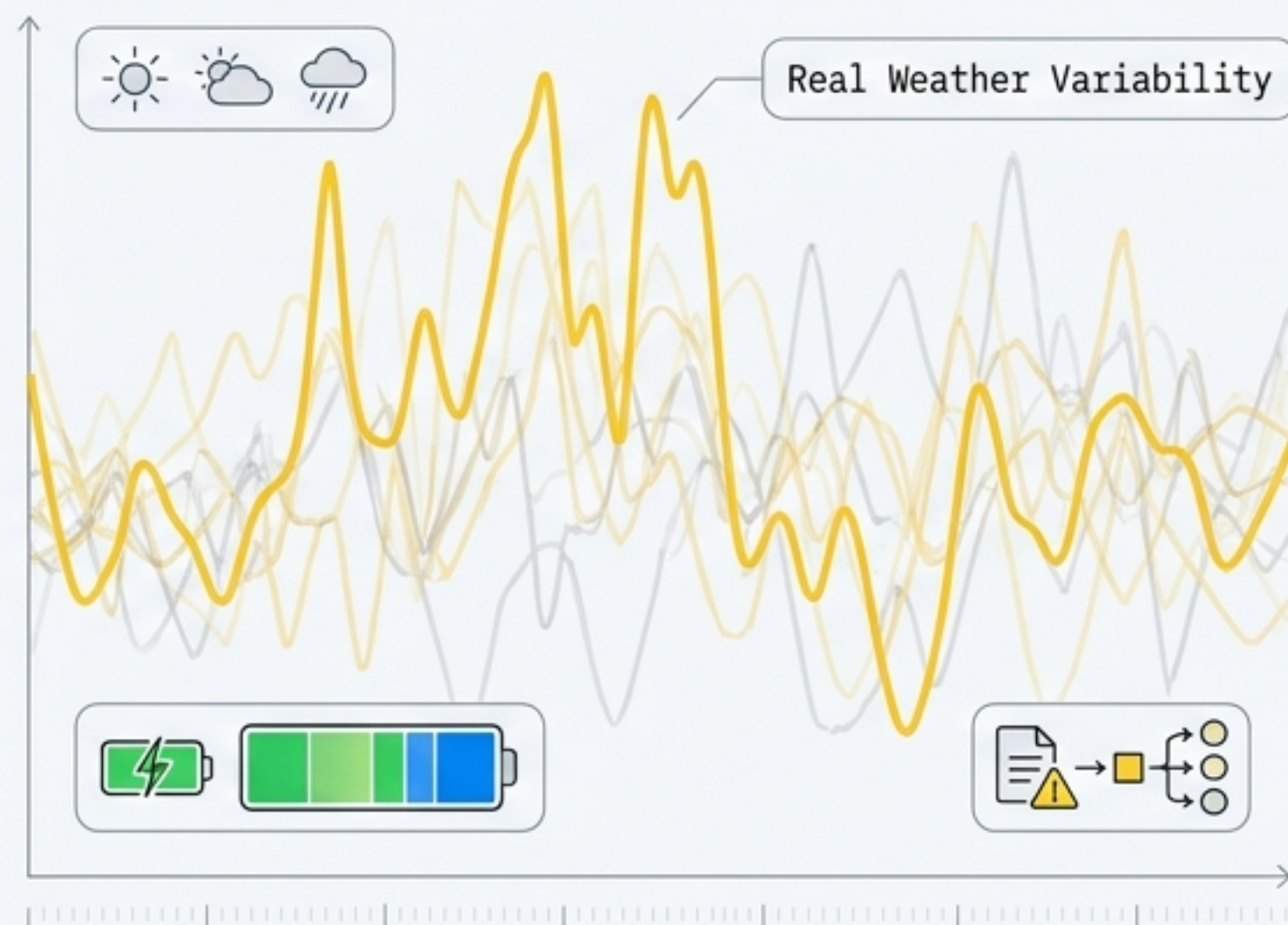
Why Average Sizing Fails



Designing for Averages.

Sizing systems based solely on average daily yield ignores real-world fluctuations, leading to frequent, unexpected energy shortages.

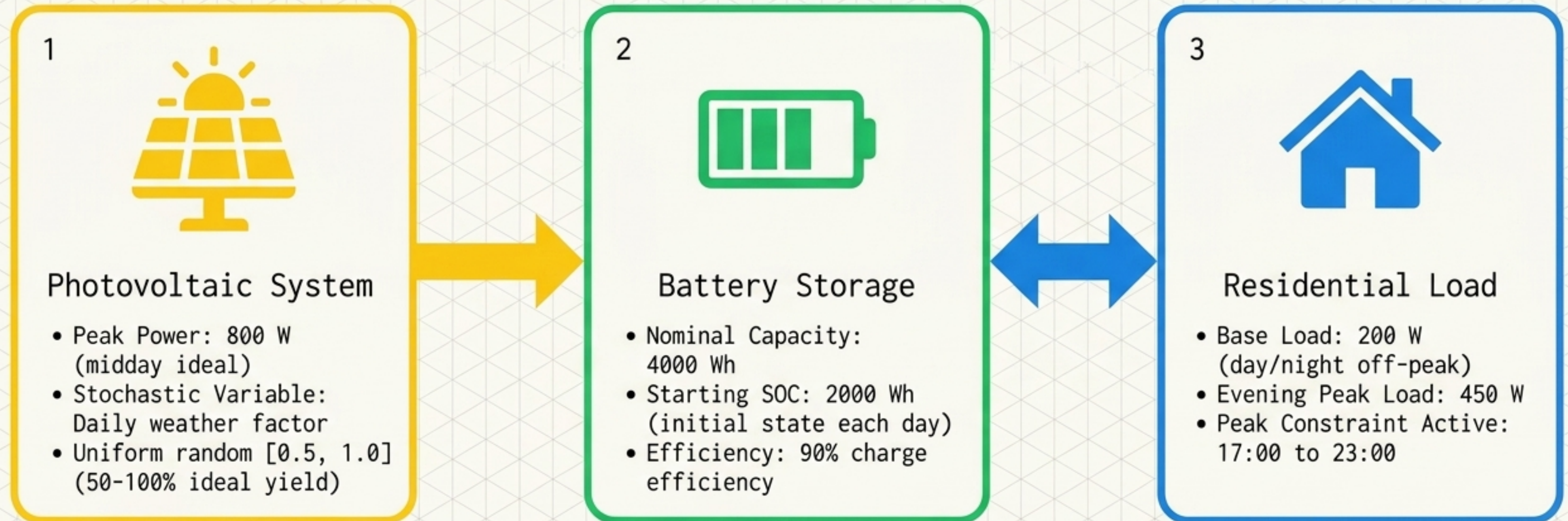
The Probabilistic Reality



Designing for Variability.

Solar generation is inherently stochastic. We must use probabilistic modeling to evaluate blackout risks, battery dynamics, and single-day autonomy under varying weather states.

System Architecture & Boundaries



Simulating the Hourly Energy Balance

Step 1: Calculate Net Energy

$$\Delta E = (P_{\text{solar}} - P_{\text{load}}) \times 0.9 \text{ (charge efficiency)}$$

Note: P_{solar} is the base profile scaled by the daily weather factor.



Step 2: Update State of Charge (SOC)

$$E_{t+1} = \min(\max(E_t + \Delta E, 0), 4000)$$



4000 Wh
Capacity



Step 3: Evaluate Reliability

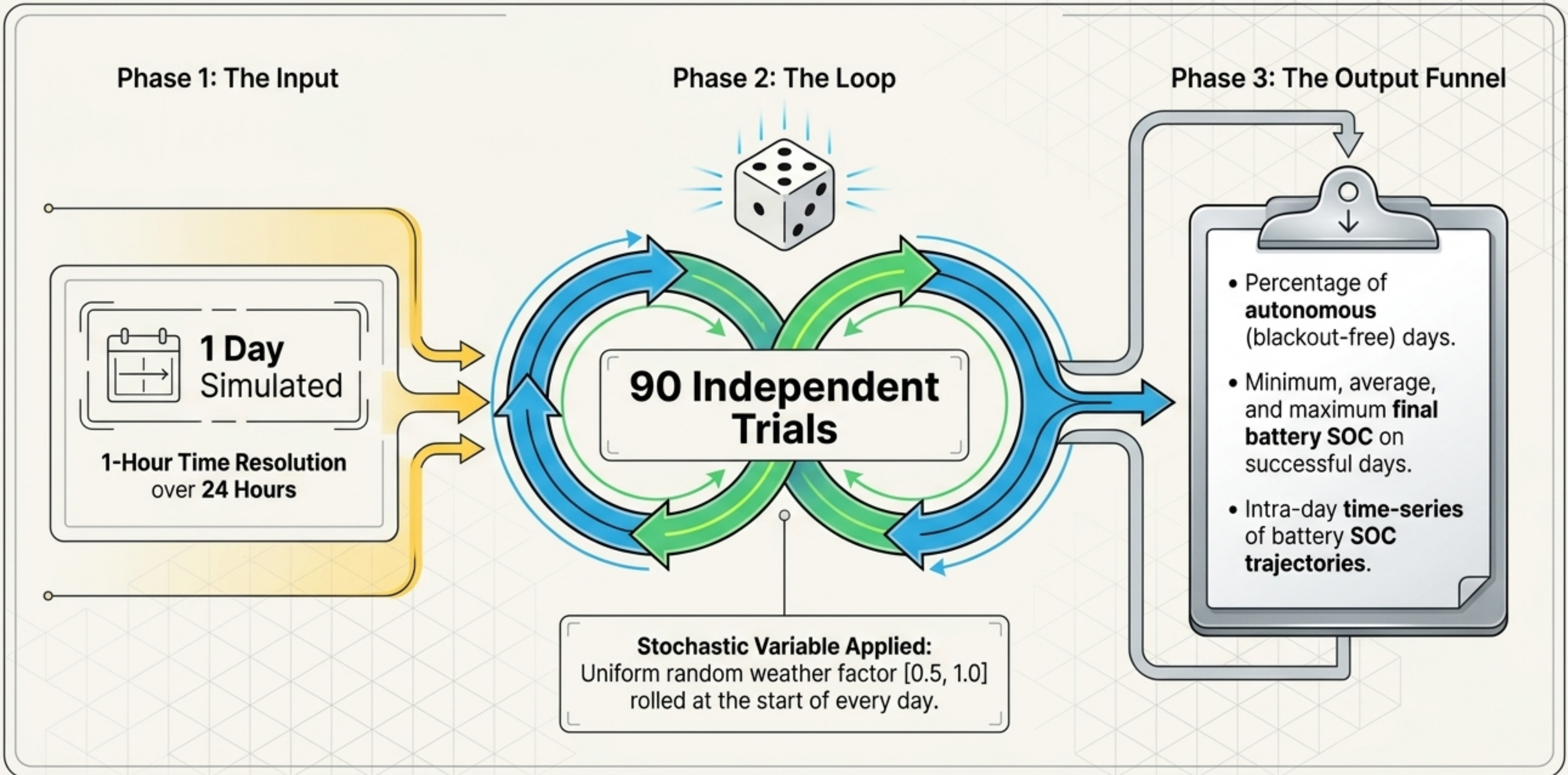
If $E_{t+1} \leq 0$ while $P_{\text{load}} > 0$



BLACKOUT STATE ⚠

Important Model Limitation: The simulation evaluates independent daily autonomy. The battery level forcibly resets to 2000 Wh at the start of each simulated day, excluding multi-day carry-over effects.

Methodology: The Monte Carlo Approach



Reliability KPIs: Monte Carlo Simulation Results

97 – 99.5%

Daily Autonomy Reliability

Zero blackouts on typically 88–90 of the simulated days.

0 – 2

Typical Blackout Days (out of 90)

Blackouts are rare (~0.5–3%).

~3200 – 3500 Wh

Average Final Capacity

Mean battery level at midnight on successful days.

~1500 – 2200 Wh

Minimum Final Capacity

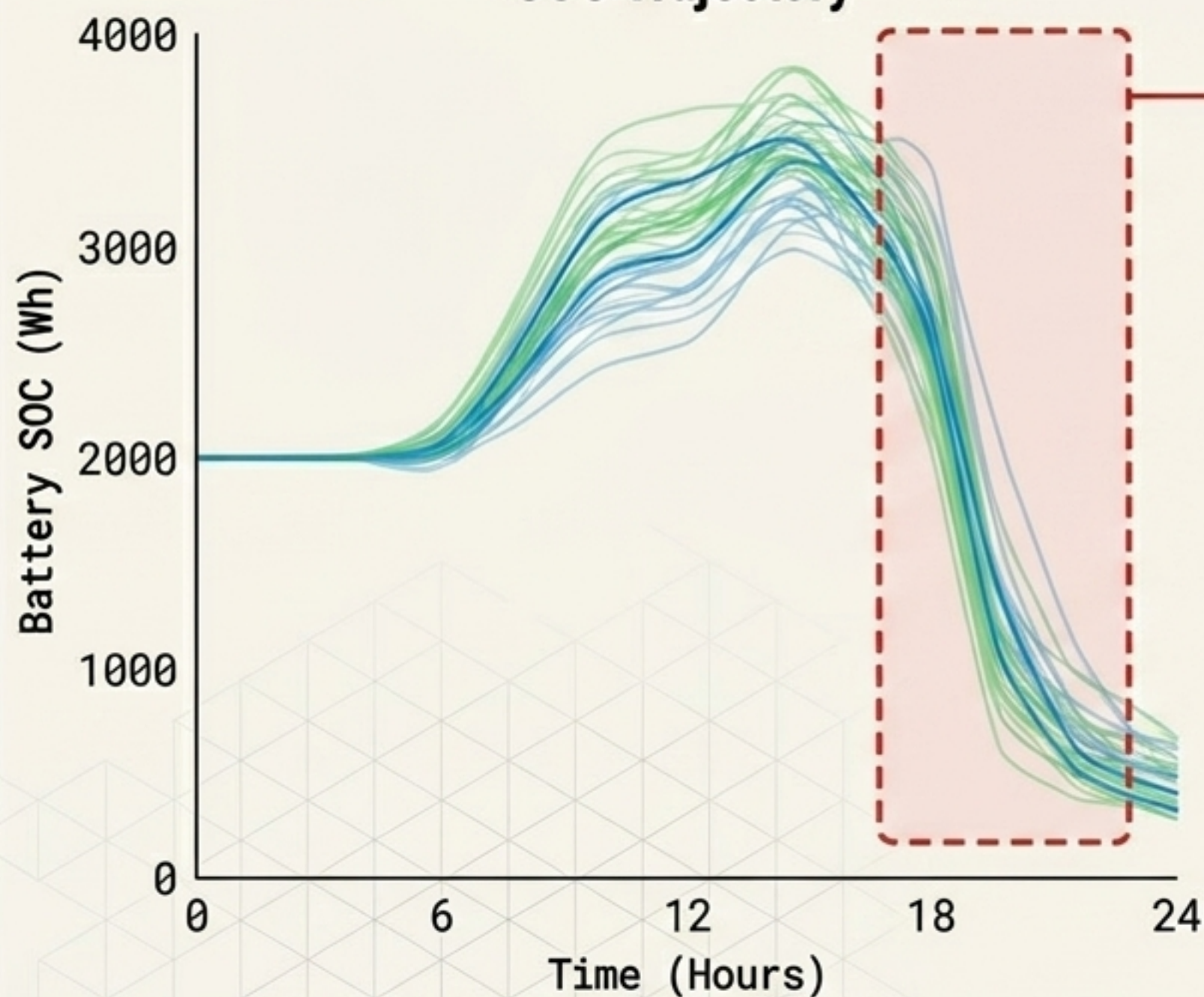
Lowest midnight battery level on successful days.

Core Conclusion: The 800W / 4000Wh configuration provides excellent **single-day autonomy**, **surviving even when solar yield plummets to 50%** under worst-case weather.

Visualizing the Dynamics

The Vulnerability Window

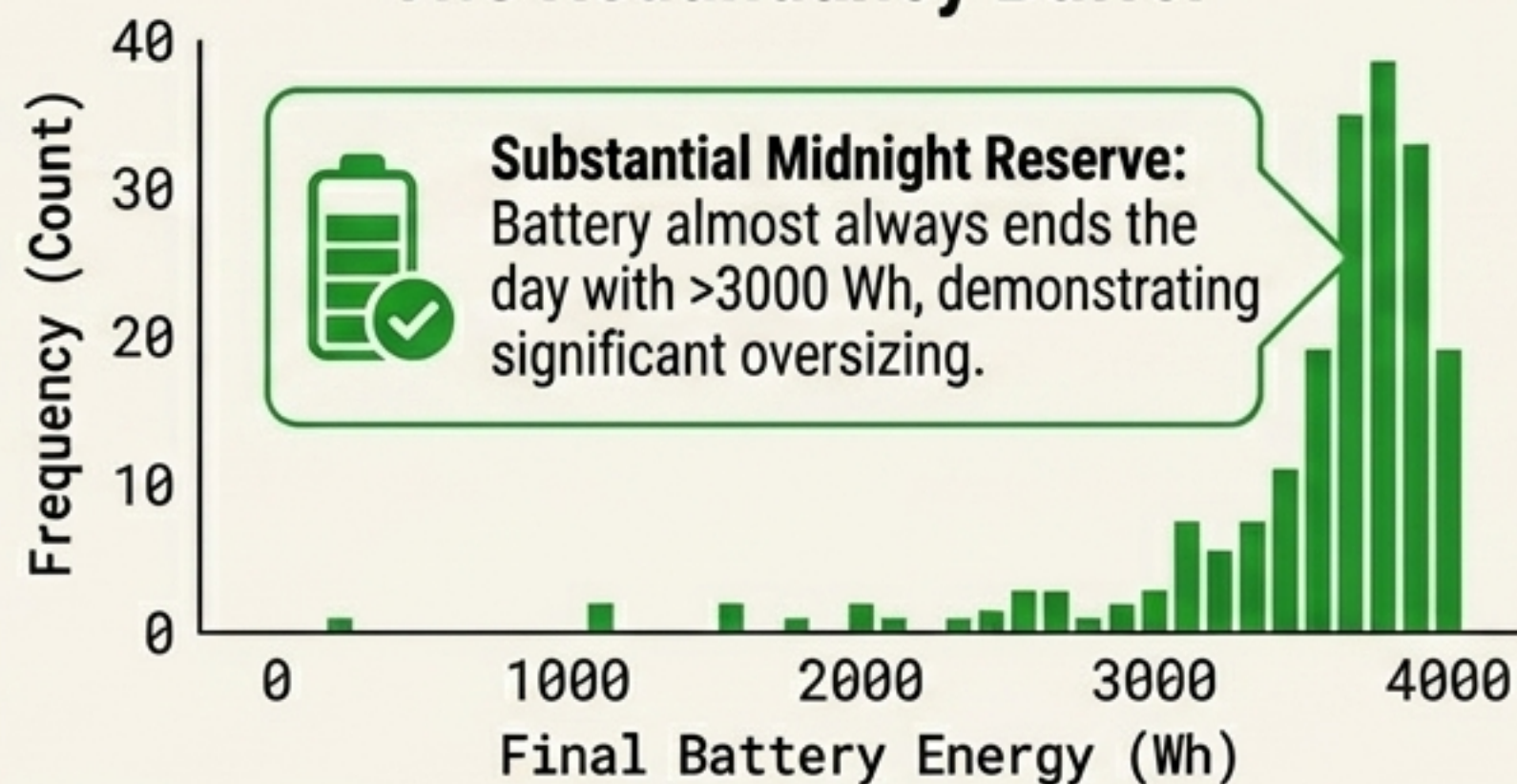
SOC Trajectory



Evening Peak Stress:

Sharp SOC drop between **17:00–23:00** caused by **450W** peak load combined with zero solar generation.

The Redundancy Buffer



Synthesis: The System Tension

High Reliability

Blackouts occur strictly under the worst-case weather permutation paired with the full evening peak. The system rarely fails.



Excessive Redundancy

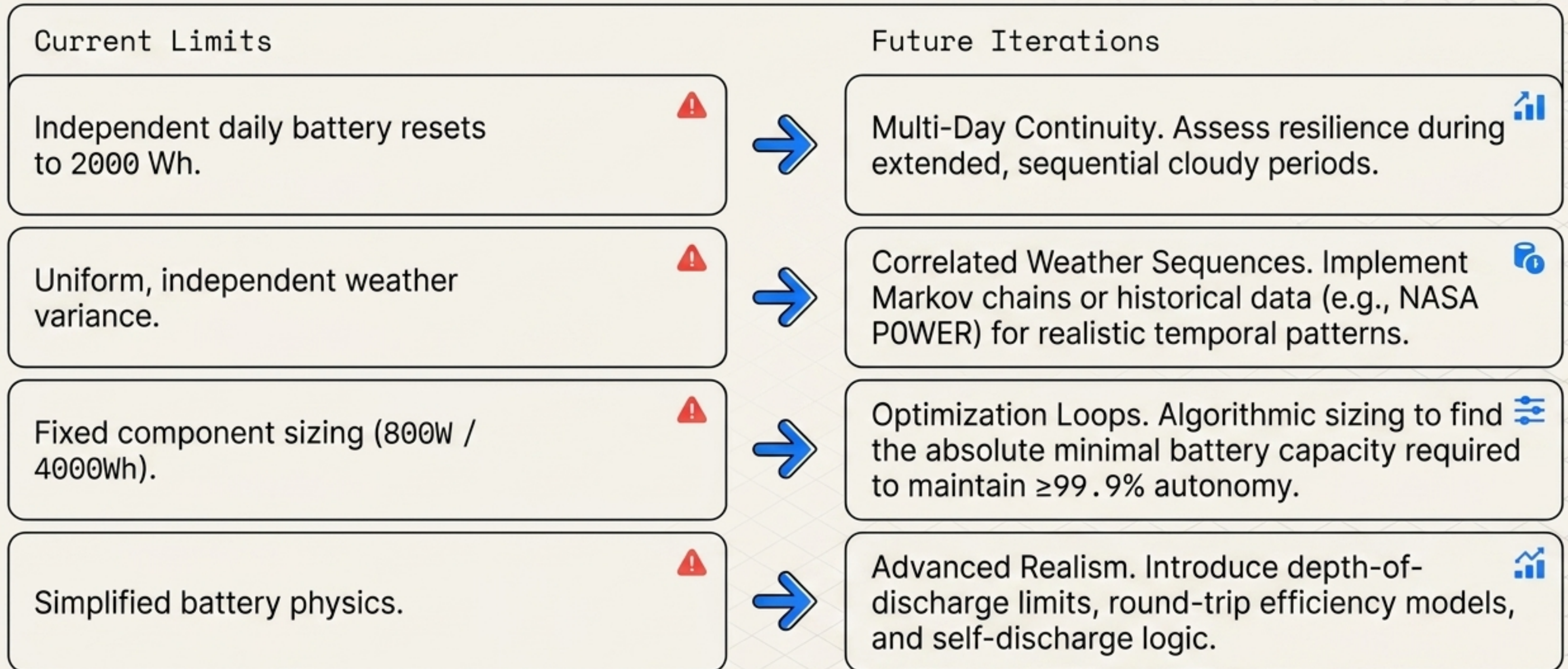
Ending most days with over 3000Wh of reserve means the 4000Wh battery is significantly oversized for single-day autonomy. Capital is tied up in unutilized capacity.



Strategic Load Shifting

The 17:00–23:00 evening window is the single point of failure. Introducing demand-side management to shift this load to peak daylight hours is the lowest-cost method to maintain 99% autonomy while allowing for a smaller battery footprint.

Horizons: Modeling Limitations & Next Steps



References: Bukar et al. (2019) Optimal sizing of microgrids; Maleki & Askarzadeh (2014); Khatib et al. (2013).